THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains why one factor is changed.
2	**(c)** A is true but R is false.	A is true; R is false because animal cells lack a cell wall.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(d)** greater	12 degrees C versus 7 degrees C.
15	**(b)** 2 mm	2 x 1000 = 2000 micrometres = 2 mm.
16	**(c)** 70	40 + 30 = 70.
17	**(b)** B gets hotter than A	Thin wire has higher resistance.
18	**(c)** 20 N	60 - 40.
19	**(d)** 3822 Pa	Area = 3.14 x 0.01 = 0.0314 square m; 120/0.0314 = 3822 Pa.
20	**(c)** spaces between particles	Particles of one fill the gaps.
21	**(c)** 36 g, 1 : 8	4 + 32 = 36; 4:32 = 1:8.
22	**(d)** 39 per cent	64/164 x 100 = 39.
23	**(c)** 140 degrees	Initial i = 55; new i = 70; angle = 140.
24	**(a)** 15 m	3/4 = h/20.
25	**(d)** snake	Energy decreases at each step.
26	**(d)** 0.53	3400/6400.
27	**(c)** 50 degrees C	Subtract zero error of 3.
28	**(d)** stops functioning properly	Nucleus controls the cell.
29	**(a)** colds are viral and often recover by themselves	Colds go away without them.
30	**(b)** the iron rod	Iron concentrates the field.
31	**(c)** moves towards team A	Net force towards A.
32	**(b)** A is likely to be stronger	Sea supplies energy.

33	**(a)** temperature increases the motion of particles	Faster particles.
34	**(b)** homogeneous mixture	Alloy.
35	**(b)** Y	Immiscible liquids.
36	**(a)** 90 degrees to the surface (back along normal)	Incidence along the normal.
37	**(b)** about 400 times farther	Equal angular sizes.
38	**(d)** bees and flowers	Both benefit.
39	**(b)** flooding by rising sea levels	Higher sea level.
40	**(c)** A is true but R is false.	A is true; R is false, repetition checks reliability but cannot guarantee truth.
41	**(a)** Both A and R are true and R is the correct explanation of A.	Equal thickness is a controlled variable; R is the rule of a fair test.
42	**(a)** Both A and R are true and R is the correct explanation of A.	The compartments he saw were the walls of dead plant cells, which is why they looked like small rooms.
43	**(d)** A is false but R is true.	A is false because health is complete well-being; R is true.
44	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's observation is explained by the magnetic field of the current.
45	**(a)** Both A and R are true and R is the correct explanation of A.	R states exactly why friction is a contact force.
46	**(a)** Both A and R are true and R is the correct explanation of A.	The sharp edge has a smaller area, so the pressure is greater.
47	**(a)** Both A and R are true and R is the correct explanation of A.	Sugar particles occupy the inter-particle spaces of water.
48	**(a)** Both A and R are true and R is the correct explanation of A.	R is the definition of an element.
49	**(a)** Both A and R are true and R is the correct explanation of A.	Very small particles cannot scatter light, so there is no Tyndall effect.
50	**(a)** Both A and R are true and R is the correct explanation of A.	A virtual image is formed by rays that only appear to meet, so it cannot be caught on a screen.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(d)** B and C	Only light differs between B and C.
4	**(b)** 0.05 mm	5 / 100 = 0.05 mm.
5	**(d)** 75%	15000/20000 x 100 = 75%.
6	**(a)** glows dimmer	Reversed cell opposes the others.
7	**(b)** 200 N	600/3.
8	**(d)** 4000 Pa	Inversely proportional.
9	**(b)** change of state	Latent heat.
10	**(a)** 1 g and 8 g	Ratio 1 : 8.
11	**(a)** 16 per cent	20+60 = 80 g in 500 g.
12	**(b)** 65 degrees	i2 = 90 - 25.
13	**(d)** decreased	Longer shadow means a lower Sun.
14	**(b)** support all the other levels and hold the most energy	All other organisms depend on producers.
15	**(d)** 42 million cubic km	0.03 x 1.4 billion.
16	**(d)** plants without fertiliser grown in same conditions	Needed for comparison.
17	**(a)** yeast is more active at warm temperature	Warmth boosts fermentation.
18	**(c)** boiling milk	Not related to vector.
19	**(c)** the current is larger	More current.
20	**(c)** magnetic force	Acts at a distance.
21	**(b)** wind and waves are strongest near the eye wall	Danger.
22	**(b)** equal at equilibrium	Dynamic equilibrium.
23	**(a)** hydrogen gas	Metal with acid.
24	**(c)** a solid dissolves in a solid	Metals.
25	**(b)** parallel	Parallel beam.
26	**(b)** about 3 years	11 days x 3 = 33 days.
27	**(b)** parasitism	One benefits, one harmed.
28	**(d)** using public transport and saving electricity	Reduces emissions.
29	**(a)** Both A and R are true and R is the correct explanation of A.	Only testable statements can be supported or rejected.
30	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
31	**(a)** Both A and R are true and R is the correct explanation of A.	Breeding sites are removed.
32	**(c)** A is true but R is false.	A is true; R is false because a thick wire has less resistance.
33	**(c)** A is true but R is false.	A is true; R is false because mass does not depend on gravity.
34	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
35	**(c)** A is true but R is false.	A is true; R is false: heat is used to overcome forces between particles.

36	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
37	**(c)** A is true but R is false.	A is true; R is false: powder has a larger surface area.
38	**(c)** A is true but R is false.	A is true; R is false because they form virtual diminished images.
39	**(c)** A is true but R is false.	A is true; R is false because it rotates once per revolution.
40	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
41	**(a)** Both A and R are true and R is the correct explanation of A.	A higher Sun casts a shorter shadow.
42	**(a)** Both A and R are true and R is the correct explanation of A.	Making food from simple substances is what makes them producers.
43	**(a)** Both A and R are true and R is the correct explanation of A.	Molten metal makes the outer core liquid.
44	**(a)** Both A and R are true and R is the correct explanation of A.	R is the definition that explains why A holds.
45	**(a)** Both A and R are true and R is the correct explanation of A.	The cell wall gives the plant cell its fixed shape.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes how the mosquito becomes infected, and A states the route to a new person, so R does not directly explain A.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the loss of magnetism on switching off does not explain why the magnet is stronger.
48	**(a)** Both A and R are true and R is the correct explanation of A.	Mass is constant because the quantity of matter is the same; only weight changes.
49	**(a)** Both A and R are true and R is the correct explanation of A.	Greater pressure at the bottom needs a stronger and thicker wall.
50	**(a)** Both A and R are true and R is the correct explanation of A.	Faster particle motion increases diffusion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 8 m	Boy is 5 m from the mirror; image 5 m behind; man at 3 m; total 8.
2	**(c)** high in the sky	Lunar eclipse occurs on a full moon, which is high in the sky at midnight.
3	**(a)** increase	Fewer predators.
4	**(a)** pressure raises the melting point	High pressure keeps it solid.
5	**(a)** time for wax to drop	Result measured.
6	**(c)** tissues	Tissue means many cells.
7	**(b)** isolating the patients and hygiene	Break transmission.
8	**(a)** retains magnetism	Steel is used for permanent magnets.
9	**(b)** apply a lubricant	Reduces friction.
10	**(c)** reduce the force of the surge	Wave barrier.
11	**(b)** oxygen	Gas.
12	**(d)** silver	Silver tarnish.
13	**(b)** dissolved solids	Residue.
14	**(c)** an erect, diminished image that covers a wide field	A convex mirror gives a wide field of view.
15	**(a)** one year	33 x 11 = 363 days.
16	**(c)** algae and fungi	Mutual benefit.
17	**(d)** the water table falls	Overdraft.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Honest recording of all data is needed.
19	**(a)** Both A and R are true and R is the correct explanation of A.	The space allows more haemoglobin.
20	**(d)** A is false but R is true.	A is false: diabetes is non-communicable. R is true: it is not caused by a pathogen.
21	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's finding.
22	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
23	**(a)** Both A and R are true and R is the correct explanation of A.	Smaller area gives higher pressure.
24	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
25	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but R only restates A as an exception.
26	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
27	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
28	**(c)** A is true but R is false.	A is true; R is false: it reflects sunlight.
29	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
30	**(c)** A is true but R is false.	A is true; R is false: the Moon has almost no atmosphere.
31	**(d)** 1.6 s	48 / 30 = 1.6 s.
32	**(c)** virus, bacterium, red blood cell	Smaller to larger.

33	**(a)** typhoid	Faecal contamination.
34	**(a)** melt	Total 7 A exceeds 5 A.
35	**(c)** 3 N to the right	Net 3 N to the left, so balance with 3 N right.
36	**(d)** 0.02 square metre	800 / 40000.
37	**(d)** water evaporates and takes heat from the air	Evaporation.
38	**(a)** $3O_2$	Coefficient before formula.
39	**(a)** 300 g	40/0.1 = 400 total; add 300.
40	**(d)** 1:50	12:00 - 10:10.
41	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but malleability and ductility do not explain conduction.
42	**(a)** Both A and R are true and R is the correct explanation of A.	Solubility of a gas falls when pressure falls.
43	**(a)** Both A and R are true and R is the correct explanation of A.	The wide view lets the driver see more traffic behind.
44	**(a)** Both A and R are true and R is the correct explanation of A.	Equal rotation and revolution periods keep one face towards us.
45	**(a)** Both A and R are true and R is the correct explanation of A.	Nutrient recycling depends on decomposers.
46	**(a)** Both A and R are true and R is the correct explanation of A.	Water vapour and air are concentrated in the lowest layer.
47	**(d)** A is false but R is true.	A is false because repeating readings improves reliability; R is true.
48	**(d)** A is false but R is true.	Animal cells do not have chloroplasts, so A is false; R is true.
49	**(d)** A is false but R is true.	Diabetes is non-communicable, so A is false; R is the correct definition of communicable diseases.
50	**(d)** A is false but R is true.	A fuse wire has a low melting point, so A is false; R is true.

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